SECURE PASSWORD MANAGEMENT SYSTEM

PRG800 – Milestone 2

Submitted to:
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Submitted by:
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1. Executive Summary

This Milestone 2 report presents the project plan for the proposed SecureVault – Secure Password Management System. The project focuses on developing a cybersecurity-focused software application designed to improve password management and credential protection for individual users and students.

In today's digital environment, users manage multiple online accounts across educational, personal, and business platforms. Many users continue to practice unsafe password habits such as password reuse, weak password creation, and insecure credential storage methods. These poor security practices increase the risk of unauthorized access, credential theft, and cybersecurity breaches.

The SecureVault system aims to address these challenges by providing secure authentication, encrypted password storage, password generation, password health analysis, and secure credential management within a centralized platform. Additional security-focused features such as two-factor authentication, secure file storage, and audit logging will further strengthen user protection and digital security awareness.

This report outlines the selected industry, defines the business problem, presents the requested system, evaluates project feasibility, and provides a structured project plan that will guide the successful development of a working software solution by the end of the academic term.

2. Introduction

This milestone outlines the planning and analysis phase for the SecureVault – Secure Password Management System project. The report defines the project scope, industry relevance, system feasibility, project requirements, and development approach that will guide the implementation of the software solution.

The SecureVault system is intended to provide users with a secure platform for storing, organizing, generating, and protecting passwords and sensitive credentials using modern cybersecurity principles. The project focuses heavily on secure software engineering concepts including encryption, authentication, password hashing, secure storage, access control, and cybersecurity best practices.

The project also supports the application of project management methodologies such as Agile development, iterative testing, collaborative planning, risk management, and software documentation. Through this project, the team will combine technical software development knowledge with practical cybersecurity and project planning skills to deliver a secure and user-friendly credential management application.

3. Industry Analysis

3.1 Project Identification

The project selected by the group is the SecureVault – Secure Password Management System. The project focuses on the cybersecurity and secure software development industry, specifically within password security and credential management solutions.

The system will help users securely manage passwords and sensitive information using encrypted storage and secure authentication methods. The application will include tools that promote stronger password hygiene, safer credential management practices, and improved digital security awareness.

The proposed system aims to reduce cybersecurity risks associated with weak passwords, password reuse, unsafe storage practices, and unauthorized access to sensitive information.

3.2 Industry Vertical (NAICS)

The selected project belongs primarily to the following North American Industry Classification System (NAICS) category:

NAICS 541512 – Computer Systems Design Services

This industry includes organizations involved in:

- software design,
- cybersecurity applications,
- information systems,
- authentication systems,
- and secure software solutions.

The project also aligns with broader sectors involving:

- cybersecurity services,
- cloud computing,

- digital identity protection,
- and secure information management systems.

3.3 Industry Explanation

The cybersecurity and secure software development industry plays a major role in the Canadian digital economy. As organizations and individuals increasingly rely on online services, digital platforms, cloud systems, and mobile applications, the importance of cybersecurity solutions continues to grow significantly.

Password management and digital identity protection systems are critical because weak password practices remain one of the leading causes of cybersecurity incidents and unauthorized access. Many users continue to store passwords insecurely or reuse passwords across multiple platforms, creating vulnerabilities that can lead to credential theft and data breaches.

Canada continues to invest heavily in cybersecurity infrastructure and digital transformation initiatives. Educational institutions, businesses, healthcare systems, and government agencies increasingly require stronger cybersecurity protections to secure sensitive information and user identities. The rapid growth of remote work, online learning, e-commerce, and cloud-based systems has further increased the demand for secure authentication and credential management technologies.

Secure password management systems contribute to improved cybersecurity awareness, safer digital practices, and reduced security risks for users operating in modern digital environments.

3.4 Team Motivation

The team selected this project because it strongly aligns with the group's interests and technical backgrounds in cybersecurity, software engineering, networking, cloud computing, database systems, and secure software development.

Several team members possess experience or interests in:

web development,

- DevOps,
- QA testing,
- cloud infrastructure,
- networking,
- database management,
- and secure application development.

The project provides an opportunity to apply these technical skills within a practical cybersecurity solution that solves a real-world digital security problem. The team believes that password security and digital identity protection are increasingly important in today's technology-driven environment, making this project both professionally valuable and technically relevant.

4. Problem Statement

The problem of insecure password management affects individual users and students. The impact of which is increased cybersecurity risks, unauthorized access, password reuse, weak authentication practices, and unsafe credential storage. And can be solved with a solution which provides secure encrypted password management, centralized credential storage, password generation, secure authentication, and password protection features.

5. System Request

5.1 System Request Description

The system requested is the SecureVault – Secure Password Management System. The system is designed for individual users and students who require secure password storage and digital credential protection.

The primary business function of the system falls under:

- cybersecurity management,
- secure credential management,
- and digital identity protection.

The software will function as a Password Management System that enables users to:

- securely store passwords,
- generate strong passwords,
- organize credentials,
- monitor password health,
- and protect sensitive files.

The purpose of the system is to replace unsafe password storage methods such as unsecured notes, spreadsheets, browsers, or repeated password reuse with a centralized secure credential management solution.

5.2 SWOT Analysis of Current Business Practices

Strengths	Weaknesses
• Users already understand the importance of passwords • Increasing awareness of cybersecurity risks • Growing adoption of digital services and online platforms • Existing technologies support secure authentication systems	• Many users reuse weak passwords • Passwords are often stored insecurely • Users frequently forget credentials • Lack of centralized password management solutions for many individuals
Opportunities	**Threats**
• Growing cybersecurity awareness globally • Increased demand for secure digital solutions • Expansion into cloud-based credential management • Potential future integration with mobile and enterprise systems	• Cyberattacks and credential theft • Data breaches • Rapidly evolving cybersecurity threats • Competition from established password management platforms

5.3 Feasibility Analysis

5.3.1 Technical Feasibility

The system is technically feasible because it will be developed using modern web technologies and cybersecurity tools that are widely accessible and supported. Team members possess technical knowledge in software development, networking, cloud systems, database design, DevOps, and QA testing.

The proposed technologies include:

- Node.js
- Express.js
- MySQL or MongoDB
- GitHub
- Encryption libraries
- Authentication frameworks

No specialized hardware will be required for development or testing.

5.3.2 Operational Feasibility

The system is operationally feasible because users increasingly require secure password management solutions. The application will be designed with a user-friendly interface and simple navigation to ensure ease of use for users with varying technical experience.

The system will provide:

- secure authentication,
- organized password management,
- and centralized credential access, while minimizing complexity for end users.

5.3.3 Economic Feasibility

The proposed solution is economically feasible because development costs are low and rely mainly on educational and open-source technologies. The system will be developed within an academic environment using freely available development tools and software frameworks.

The project also offers long-term value by promoting improved cybersecurity awareness and reducing risks associated with credential compromise and unauthorized access.

5.3.4 Schedule Feasibility

The project is achievable within the academic semester timeline because the scope is well-defined and development will follow an Agile/Scrum methodology. The iterative development approach allows the team to progressively implement and test features throughout the term.

Regular sprint reviews, task planning, and milestone tracking will support timely project completion.

6. Project Plan

6.1 Project Charter

Project Name	SecureVault – Secure Password Management System
Project #	PRG800 – SecureVault System

Initiator / Client	Course Project (Academic Client)	**Department**	Information Technology
Sponsor	Professor Miles McDonald	**Division**	PRG800
Project Manager	Harmon Anthony Tuazon	**Project Size**	Medium
		Project Type	Academic / Cybersecurity Software Project

DOCUMENT REVISION MANAGEMENT		
Date	**Description of Revision**	**Revised By**
2026-05-14	Draft Project Charter	Project Team
2026-05-26	Final Project Charter	Project Team

PROJECT PURPOSE AND OBJECTIVE(S)
The purpose of this project is to design and develop a Secure Password Management System focused on improving password security, credential protection, and secure authentication practices for users operating in digital environments. Modern users manage multiple online accounts across educational, personal, business, and cloud-based platforms. Many users continue to use weak passwords, reuse credentials across systems, or store passwords insecurely using browsers, notes, spreadsheets, or unsecured applications. These practices significantly increase the risks of unauthorized access, credential theft, identity compromise, and cybersecurity breaches. This project aims to address common cybersecurity challenges such as weak password practices, unsafe credential storage, poor password hygiene, and lack of centralized password management. The system will support users by providing secure encrypted password storage, password generation, password health analysis, secure authentication, Two-Factor Authentication (2FA), and centralized credential management. The overall objective is to improve digital security awareness, strengthen credential protection, reduce cybersecurity risks, and provide users with a secure and user-friendly password management platform.

ALIGNMENT TO SECUREVAULT BUSINESS GOALS	
Strategic Pillar	**Improve Digital Security and Credential Protection**
Objectives	Improve password security and authentication practices
Tactics	Implement encrypted storage, authentication, and secure credential management
Action Item	Develop a centralized secure password management platform

PROJECT DESCRIPTION (SUMMARY-LEVEL)

The SecureVault project applies a cybersecurity-focused and user-centered approach to password management and credential protection. The system will provide users with secure tools for managing passwords and sensitive information through encryption, secure authentication, and centralized storage.

The project will focus on improving:

- password organization,
- credential protection,
- digital security awareness,
- and authentication practices.

The system will provide secure and practical cybersecurity tools that improve password hygiene while reducing cybersecurity risks associated with weak authentication practices and insecure storage methods.

CORE THEMES

Cybersecurity and Credential Protection

The SecureVault system will provide:

- encrypted credential storage,
- secure authentication,
- password hashing,
- password generation,
- and secure access management.

Ease of Use

The system will provide:

- simple navigation,
- user-friendly interfaces,
- organized credential management,
- and accessible security features.

Security Monitoring

The system will provide:

- password health analysis,
- audit logs,
- security alerts,
- and session protection mechanisms.

Design and Practicality

The system will provide:

- scalable architecture,
- practical cybersecurity features,
- and real-world password management functionality.

PRODUCT SCOPE	
IN Scope	**OUT of Scope**
Secure user login and registration	Banking/payment processing
Password hashing and encryption	Enterprise IAM systems
Password vault storage	AI-based threat detection
Password generator	Browser extension integration
Password strength analysis	Third-party enterprise integrations
Secure file storage	Commercial deployment
Two-Factor Authentication (2FA)	Real-time enterprise monitoring
Audit logging	Cloud enterprise infrastructure

KEY STAKEHOLDERS	
Role	**Responsibility**
Project Manager	Oversees planning and coordination
Technical Lead	Designs system structure and architecture
Developers	Implement features and functionality
QA Tester	Performs testing and validation
Instructor / Sponsor	Provides academic guidance and approval
End Users	Use the system for password management

6.2 Deliverables, Constraints, Acceptance Criteria & Risks

<table>
<tr>
<td>Deliverables
The project will deliver:
Business requirements documentation
SecureVault system design
Database and architecture design
Functional software prototype
Testing documentation
Final report and presentation</td>
<td>Project Constraints
The project is subject to:
academic milestone deadlines,
limited development time,
limited team resources,
academic scope restrictions,
and semester scheduling limitations.</td>
</tr>
<tr>
<td>Acceptance Criteria
The project will be considered successful when:
encrypted password storage functions correctly,
secure authentication is implemented,
password generation works properly,
password health analysis operates successfully,
documentation is complete and professional,
and all milestones are submitted on time.</td>
<td>Initial Risks
The project may face:
time management challenges,
feature integration issues,
evolving requirements,
cybersecurity vulnerabilities,
and uneven workload distribution.
Risk Mitigation
Risks will be managed through:
weekly meetings,
milestone tracking,
GitHub collaboration,
iterative testing,
instructor feedback,
and Agile sprint planning.</td>
</tr>
</table>

7. High-Level Project Plan

The SecureVault project will follow an Agile/Scrum development methodology. The project schedule is organized into multiple sprints covering requirements gathering, system design, software development, testing, and final project delivery activities.

The following Gantt chart presents the high-level project timeline and task scheduling structure for the SecureVault – Secure Password Management System.

7.1 High-Level Gantt Chart

ID	Task Name	Duration	Start Week	Finish Week
1	Sprint 1 – Requirements & Planning	1 Week	Week 1	Week 1
1.1	Requirements gathering	2 Days	Week 1	Week 1
1.2	Problem analysis	1 Day	Week 1	Week 1
1.3	Feature definition	1 Day	Week 1	Week 1
1.4	Industry research	1 Day	Week 1	Week 1
2	Sprint 2 – System Design	1 Week	Week 2	Week 2
2.1	UI/UX design	2 Days	Week 2	Week 2
2.2	Database design	2 Days	Week 2	Week 2
2.3	Architecture planning	1 Day	Week 2	Week 2
3	Sprint 3 – Core Development	2 Weeks	Week 3	Week 4
3.1	User authentication	3 Days	Week 3	Week 3
3.2	Encryption implementation	3 Days	Week 3	Week 3
3.3	Password vault development	1 Week	Week 4	Week 4
4	Sprint 4 – Advanced Features	2 Weeks	Week 4	Week 5
4.1	Password generator	2 Days	Week 4	Week 4
4.2	Password health dashboard	2 Days	Week 4	Week 4
4.3	Secure file storage	3 Days	Week 5	Week 5
4.4	2FA implementation	2 Days	Week 5	Week 5
5	Sprint 5 – Testing & QA	1 Week	Week 6	Week 6
5.1	Functional testing	2 Days	Week 6	Week 6
5.2	Security testing	2 Days	Week 6	Week 6
5.3	Bug fixing	1 Day	Week 6	Week 6
5.4	Validation	1 Day	Week 6	Week 6
6	Sprint 6 – Finalization	1 Week	Week 7	Week 7

ID	Task Name	Duration	Start Week	Finish Week
6.1	Documentation	2 Days	Week 7	Week 7
6.2	Final deployment	1 Day	Week 7	Week 7
6.3	Presentation preparation	1 Day	Week 7	Week 7
6.4	Submission	1 Day	Week 7	Week 7

Table 7.1: High-Level Agile/Scrum Gantt Chart Schedule for SecureVault Project

Description:
This table presents the project work breakdown schedule, sprint activities, task durations, and planned completion timeline for the SecureVault project.

7.2 Project Milestones

Milestone ID	Milestone	Expected Completion
M1	Requirements Gathering Completed	Week 1
M2	System Design Completed	Week 2
M3	Core Development Completed	Week 4
M4	Advanced Features Completed	Week 5
M5	Testing & Quality Assurance Completed	Week 6
M6	Final Documentation and Submission	Week 7

Table 7.2: SecureVault Project Milestones

Description:
This table summarizes the major project milestones and their expected completion dates throughout the Agile/Scrum development lifecycle.

7.3 Gantt Chart Schedule Overview

The SecureVault project schedule is divided into six Agile/Scrum sprints that support iterative development, continuous testing, collaboration, and milestone tracking throughout the academic semester.

The project begins with requirements gathering, problem analysis, feature definition, and industry research activities. Following the planning stage, the project progresses into system design activities including UI/UX design, database design, and architecture planning.

The development phase focuses on implementing secure authentication, password encryption, and password vault functionality, followed by advanced cybersecurity features such as password generation, password health analysis, secure file storage, and Two-Factor Authentication (2FA).

The final project stages include functional testing, security testing, bug fixing, validation, documentation preparation, deployment activities, and final project presentation submission.

Figures 7.1 through 7.4 provide calendar-based and Gantt chart visualizations of the SecureVault project schedule, illustrating sprint activities, milestone timelines, and major project deliverables throughout the Agile/Scrum development lifecycle.

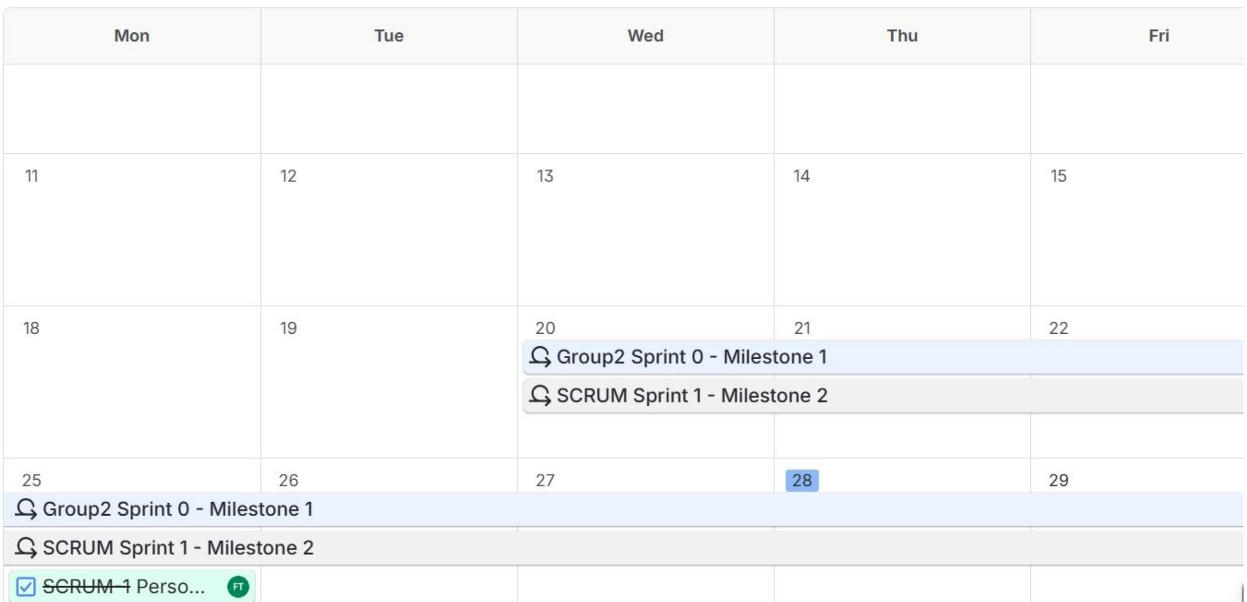

Figure 7.1: SecureVault Project Calendar Schedule – Sprint 1 and Sprint 2 Activities (May–June 2026)

Description: Calendar-based project schedule showing Sprint 1 (Requirements and Planning) and Sprint 2 (System Design) activities, including requirements gathering, problem analysis, feature definition, industry research, UI/UX design, database design, and architecture planning

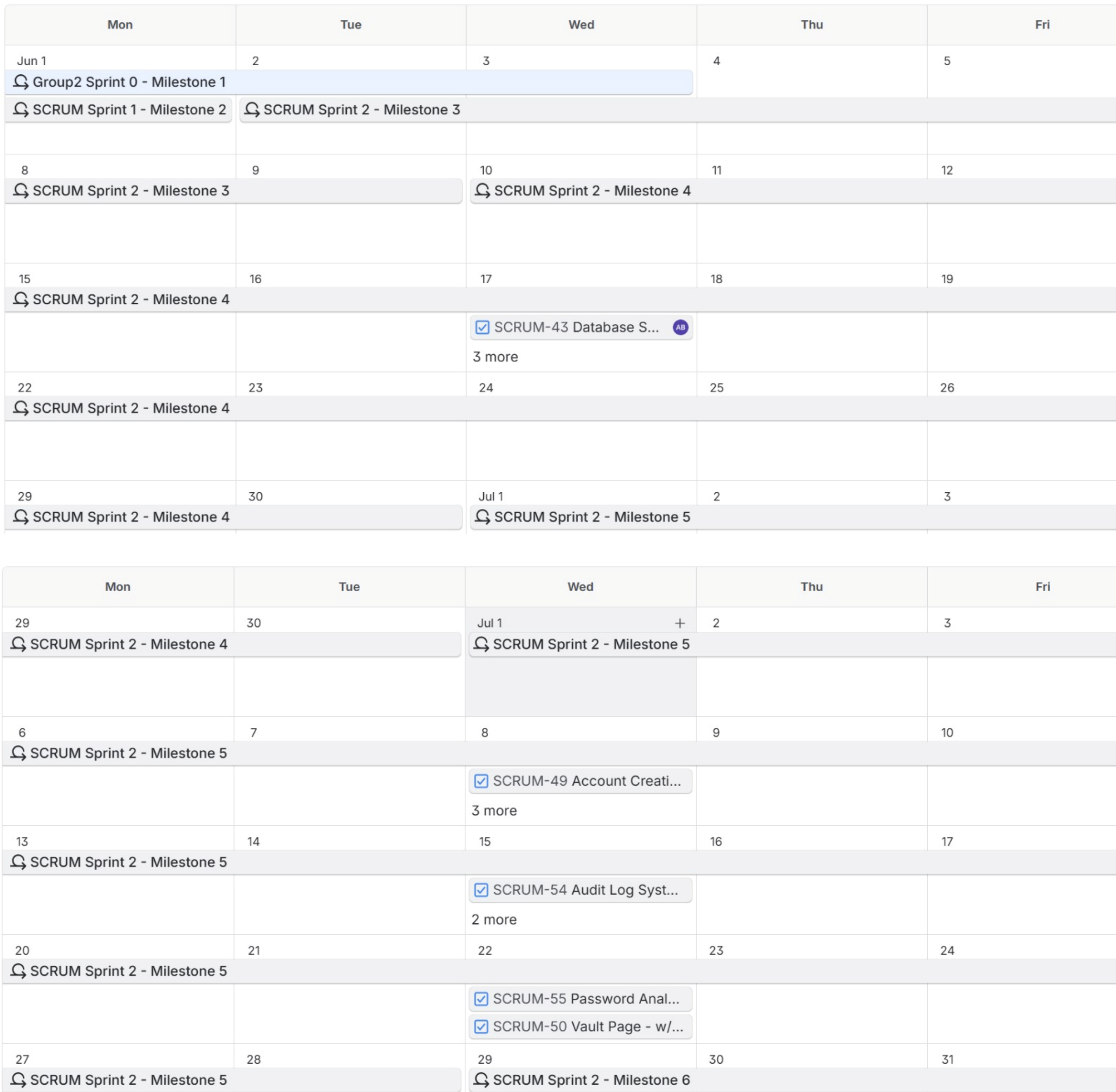

Figure 7.2: SecureVault Project Calendar Schedule – Core Development Activities (July 2026)

Description: Calendar schedule illustrating Sprint 3 activities, including user authentication development, encryption implementation, password vault development, and associated software development tasks.

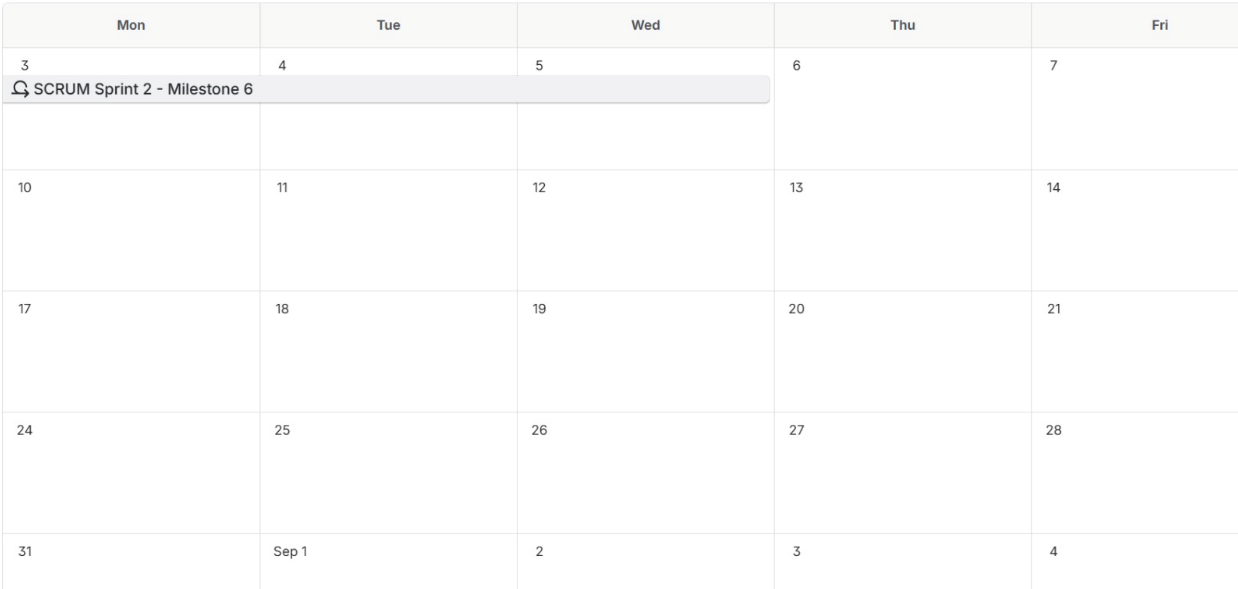

Figure 7.3: SecureVault Project Calendar Schedule – Advanced Features, Testing, and Finalization Activities (July–August 2026)

Description: Calendar schedule showing Sprint 4, Sprint 5, and Sprint 6 activities, including password generation, password health analysis, secure file storage, Two-Factor Authentication (2FA), testing, quality assurance, documentation, deployment, and final submission milestones

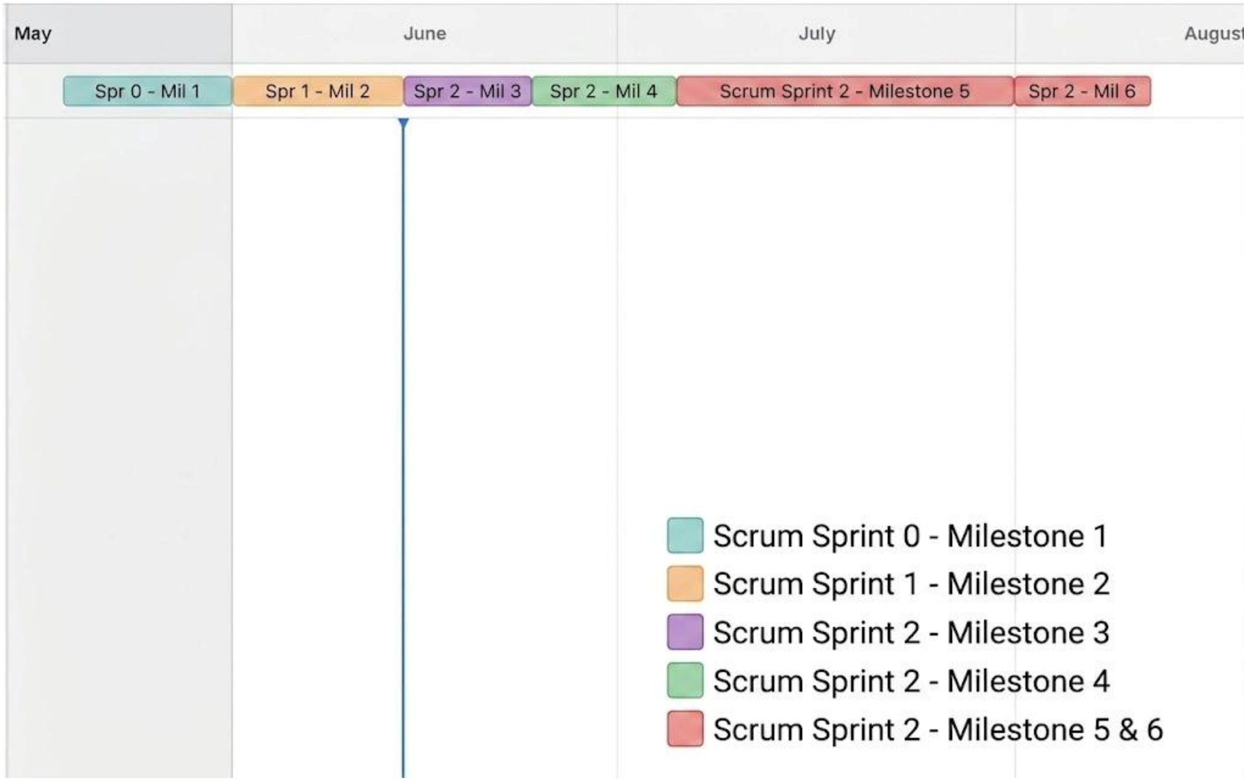

Figure 7.4: High-Level Agile/Scrum Gantt Chart for SecureVault – Secure Password Management System

Description: This figure provides a visual representation of the overall project timeline, illustrating sprint milestones, major deliverables, project phases, and milestone completion dates from project initiation through final project submission.

8. Summary

This milestone provided a comprehensive project plan for the SecureVault – Secure Password Management System. Through industry analysis, cybersecurity problem identification, feasibility evaluation, and project planning, the team has demonstrated that the proposed solution is practical, relevant, and achievable within the academic term.

The project combines cybersecurity principles, secure software engineering, and Agile project management practices to deliver a functional and secure credential management application. By focusing on password security, encrypted storage, and secure authentication, the SecureVault system aims to improve digital security practices and reduce cybersecurity risks for users.

The project scope is clearly defined, technically feasible, and aligned with both academic objectives and modern cybersecurity requirements.

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